

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 13: Dense Media and the Collapse Limit

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Overview

Hydrostatic equilibrium and radial stability are solved for a dense medium with an explicit equation of state.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$dP/dr = -\rho d\Phi/dr + f_m(r; \theta_D)$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$\omega_0^2(M_c) = 0$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$M_c = 1.9M_\odot$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- SGT9 dense-medium block D-01–D-07
- Compact Medium Objects, ch. 3
- Collapse Limit Calibration Report 2026